MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $(x^3)'$ equals
   - A. x^2
   - B. $3x^2$
   - C. $3x$
   - D. $\frac{x^4}{4}$
2. $(\frac{1}{x})'$ equals
   - A. $-\frac{1}{x^2}$
   - B. $\frac{1}{x^2}$
   - C. $-\frac{1}{x}$
   - D. 1
3. $(uv)'$ equals
   - A. $u'v'$
   - B. $u' + v'$
   - C. uv'
   - D. $u'v + uv'$
4. $f' > 0$ on an interval implies
   - A. f positive
   - B. f negative
   - C. f strictly increasing
   - D. f constant
5. $\frac{x^2+1}{x-1}$ is defined if
   - A. $x > 0$
   - B. $x \neq 1$
   - C. $x \neq -1$
   - D. $x \neq 0$

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $x^3 - x$ is
   - A. even
   - B. constant
   - C. odd
   - D. neither even nor odd
2. $f\prime(a) = 0$ necessarily implies
   - A. a horizontal tangent
   - B. a maximum
   - C. a minimum
   - D. f constant
3. Derivative of $\sqrt{x}$:
   - A. $2\sqrt{x}$
   - B. $\dfrac{1}{\sqrt{x}}$
   - C. $\dfrac{\sqrt{x}}{2}$
   - D. $\dfrac{1}{2\sqrt{x}}$, $x > 0$
4. Minimum of $x + \dfrac{1}{x}$ on $(0, +\infty)$:
   - A. 0
   - B. 2
   - C. 1
   - D. none
5. Newton's method for $x^2 - 2$ is not defined with $x_0 =$
   - A. 1
   - B. 2
   - C. 0
   - D. -1

True or false? Justify.

1. The derivative of a positive function is positive.
2. An interior extremum of a differentiable function gives a zero derivative.
3. The product of two increasing functions is increasing.

> **Key point · Self-assessment.** I differentiate on the right domain; I simplify without losing the forbidden values; I distinguish local from global extremum; I sometimes choose a factorisation rather than a derivative.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.